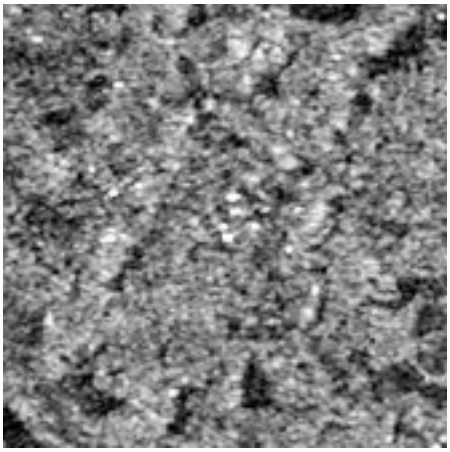
SAR Image Analysis Report



Original SAR

Colorized

K-Means Segmented

Metadata:

- Original File: original_1764392088.png
- Colorized File: colorized_1764392088.png
- Processed On: 2025-11-29 10:25:10 IST
- SAR Classifier Confidence: 0.993
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Vegetation (Green Tone) | Mean RGB: (41, 59, 46) | Area: 25.84%

Cluster 1: Likely Vegetation (Green Tone) | Mean RGB: (114, 125, 107) | Area: 25.66%

Cluster 2: Likely Vegetation (Green Tone) | Mean RGB: (74, 93, 74) | Area: 36.96%

Cluster 3: Likely Vegetation (Green Tone) | Mean RGB: (162, 168, 150) | Area: 11.55%

Land Cover Summary:

- Likely Vegetation (Green Tone) → 25.84%
- Likely Vegetation (Green Tone) → 25.66%
- Likely Vegetation (Green Tone) → 36.96%
- Likely Vegetation (Green Tone) → 11.55%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely vegetation (green tone) (100.01% combined).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is vegetation (36.93%). Strong vegetation signal suggests this area is suitable for agricultural monitoring — consider crop health and biomass monitoring. Secondary clusters show distributed variations; using time-series imagery would help confirm seasonal changes. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.993
Dominant Class Percentage: 36.96%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.